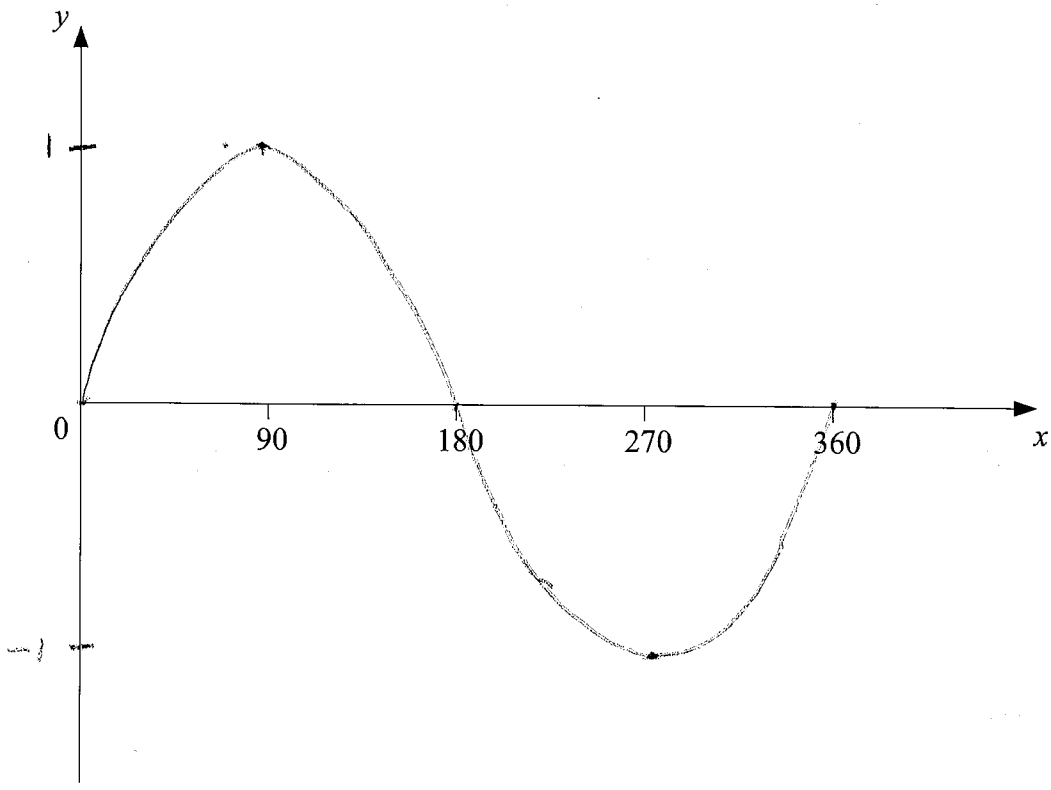
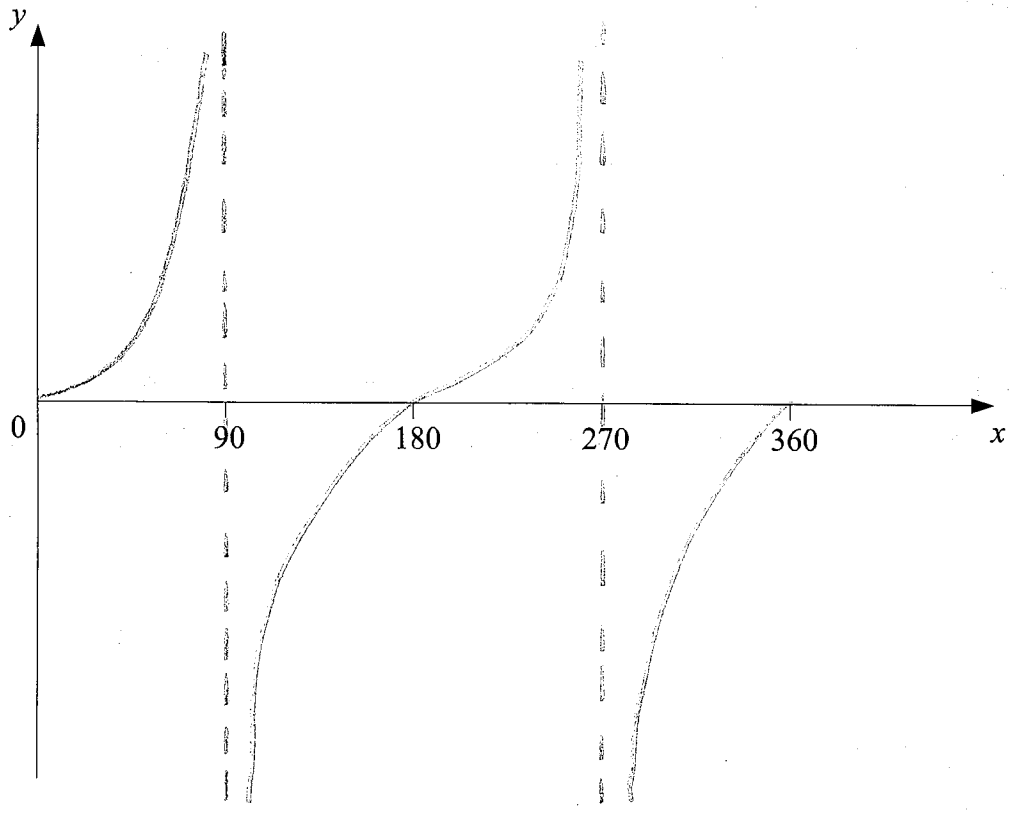
1 Sketch the graph of $y = \sin x°$ for $0 \leq x \leq 360$



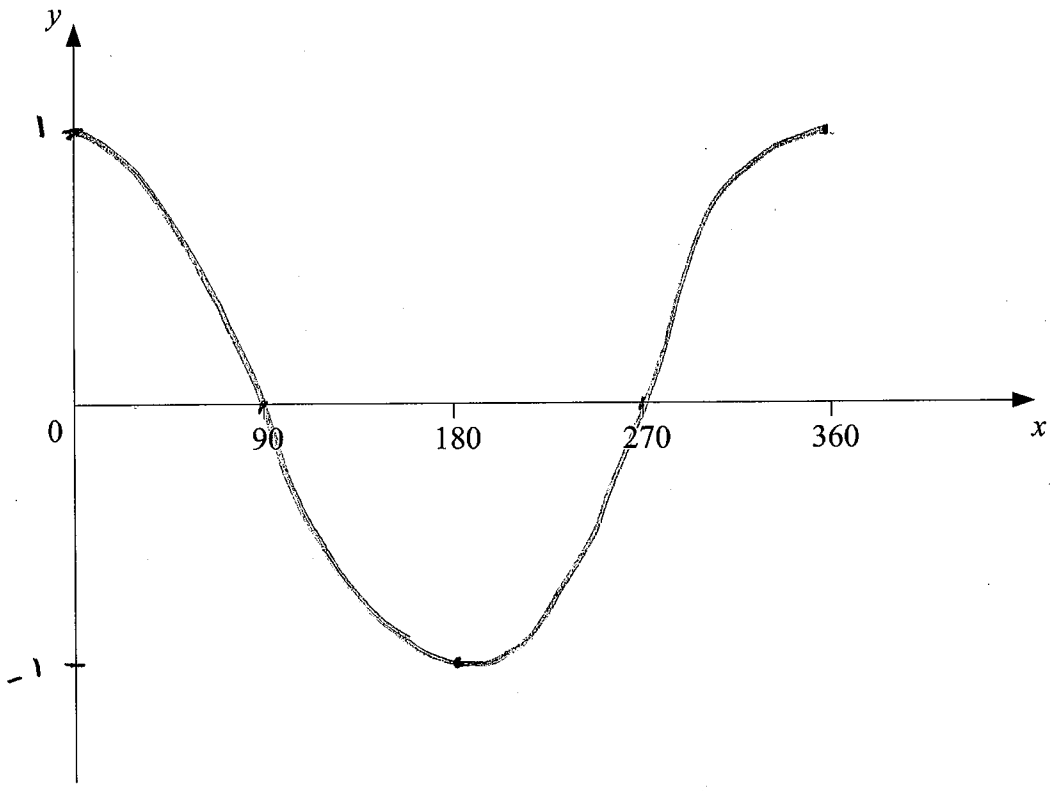
(Total for Question 1 is 2 marks)

2 Sketch the graph of $y = \tan x°$ for $0 \leq x \leq 360$



(Total for Question 2 is 2 marks)

3 Sketch the graph of $y = \cos x°$ for $0 \leq x \leq 360$



(Total for Question 3 is 2 marks)

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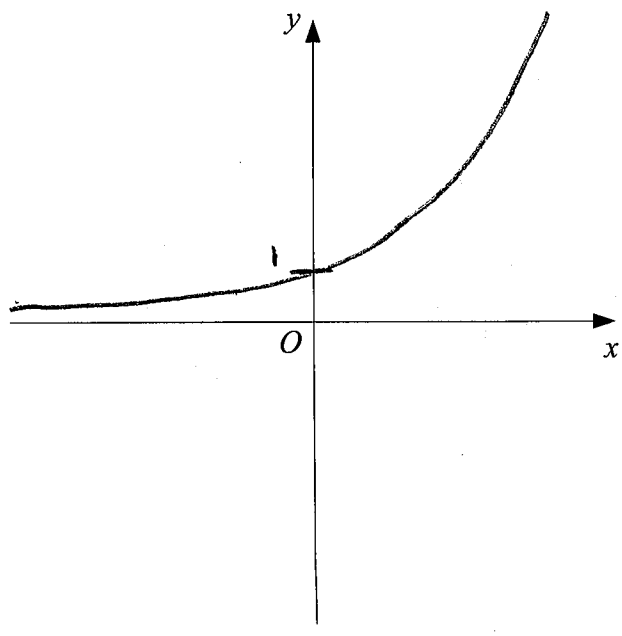
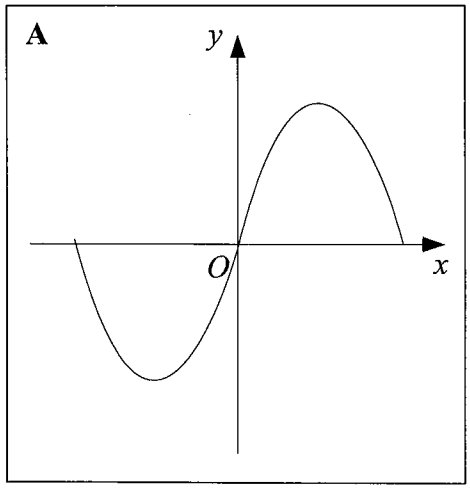
4 On the grid, sketch the curve with equation $y = 2^x$
Give the coordinates of any points of intersection with the axes.

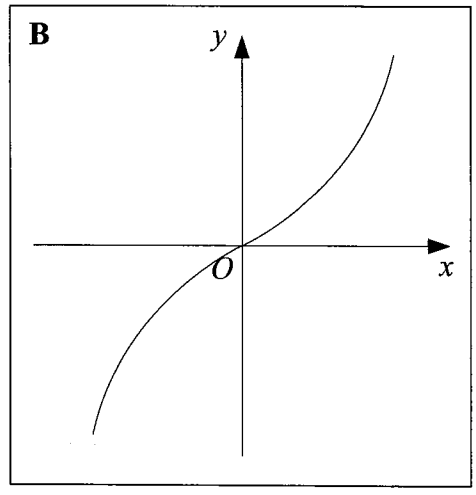


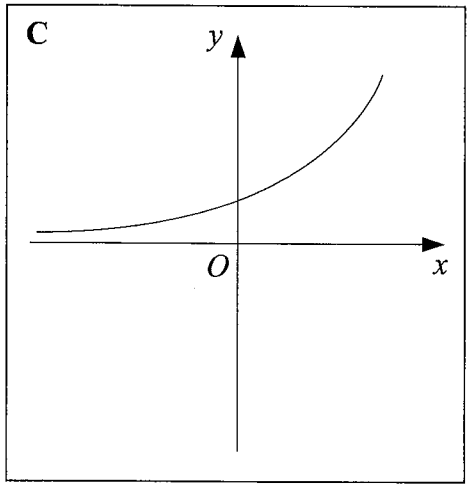
(0, 1)

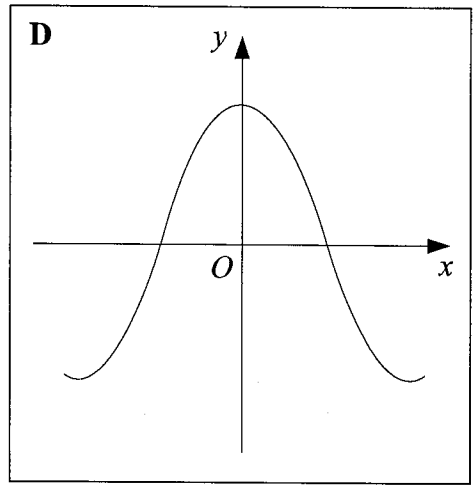
(Total for Question 4 is 2 marks)

5 Here are four graphs







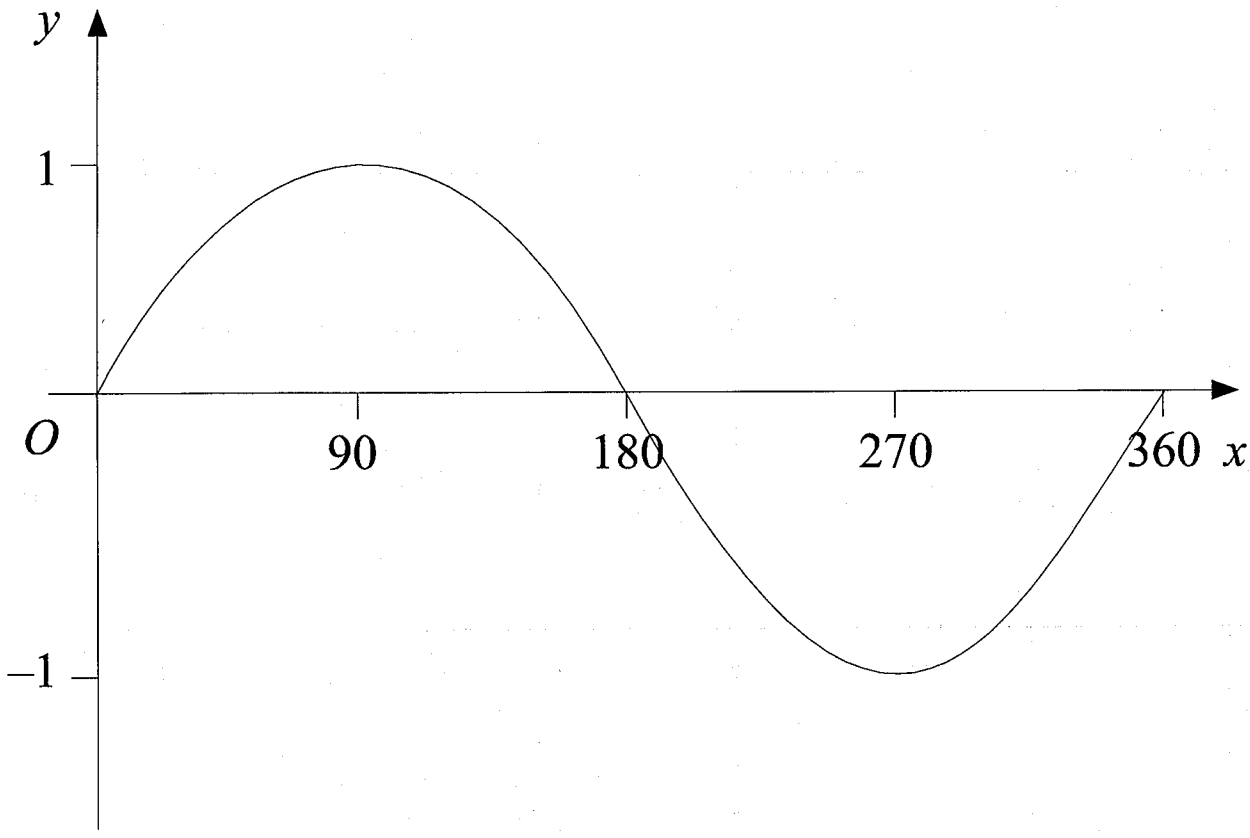


In the table below, match each equation with the letter of its graph.

Equation	Letter of Graph
$y = \sin x$	A
$y = 2^x$	C
$y = x^3$	B
$y = \cos x$	D

(Total for Question 5 is 2 marks)

6 Here is a sketch of the curve $y = \sin x°$ for $0 \leq x \leq 360$



Given that $\sin 30° = \frac{1}{2}$ write down the value of:

i) $\sin 150°$

$\frac{1}{2}$

(1)

ii) $\sin 330°$

$-\frac{1}{2}$

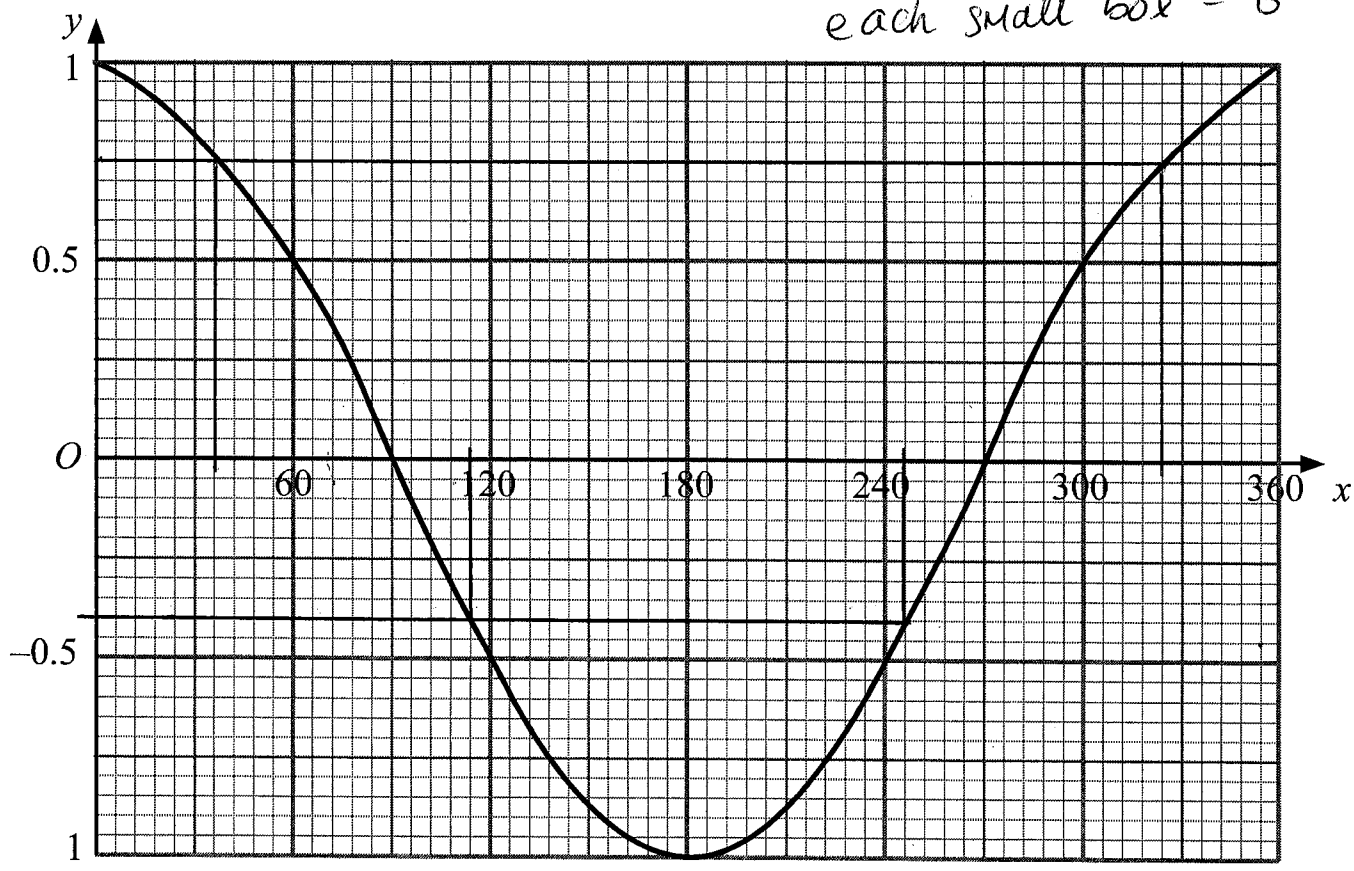
(1)

(Total for Question 6 is 2 marks)

7 Here is a sketch of the curve $y = \cos x°$ for $0 \leq x \leq 360$

$\frac{60}{10} = 6$

each small box = 6°



Use the graph to find estimates of the solutions, in the interval $0 \leq x \leq 360$, of the equation:

i) $\cos(x) = -0.4$

114° and 246°

(2)

ii) $4\cos(x) = 3$

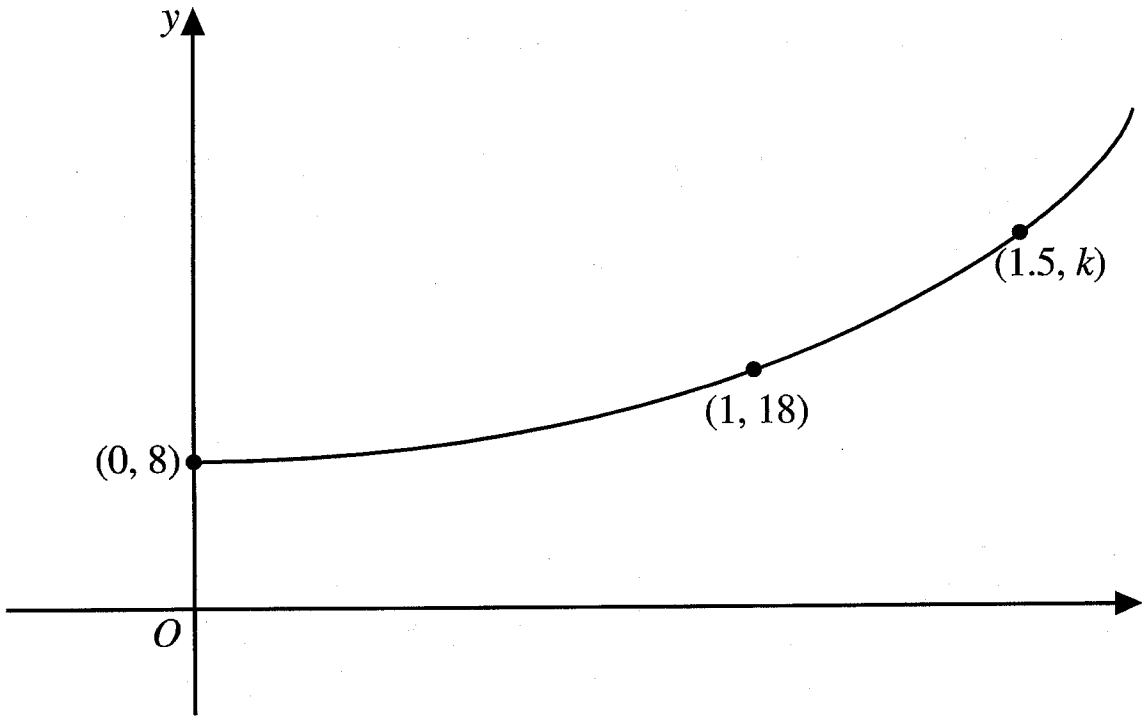
$\cos(x) = \frac{3}{4}$

36° and 324°

(2)

(Total for Question 7 is 4 marks)

8 This sketch shows part of the graph with equation $y = pq^x$ where p and q are constants.



The points with coordinates (0, 8), (1, 18) and (1.5, k) lie on the graph.

Calculate the values of p, q and k.

$y = pq^x \quad (0, 8)$

$8 = pq^0$

$\underline{\underline{8 = p}}$

$y = 8q^x \quad (1, 18)$

$18 = 8q$

$q = \frac{18}{8}$

$= \underline{\underline{\frac{9}{4}}}$

$y = 8\left(\frac{9}{4}\right)^x \quad (1.5, k)$

$k = 8\left(\frac{9}{4}\right)^{1.5}$

$= 8\left(\frac{9}{4}\right)^{\frac{3}{2}}$

$= 8\left(\frac{3}{2}\right)^3$

$= 8\left(\frac{27}{8}\right)$

$= \underline{\underline{27}}$

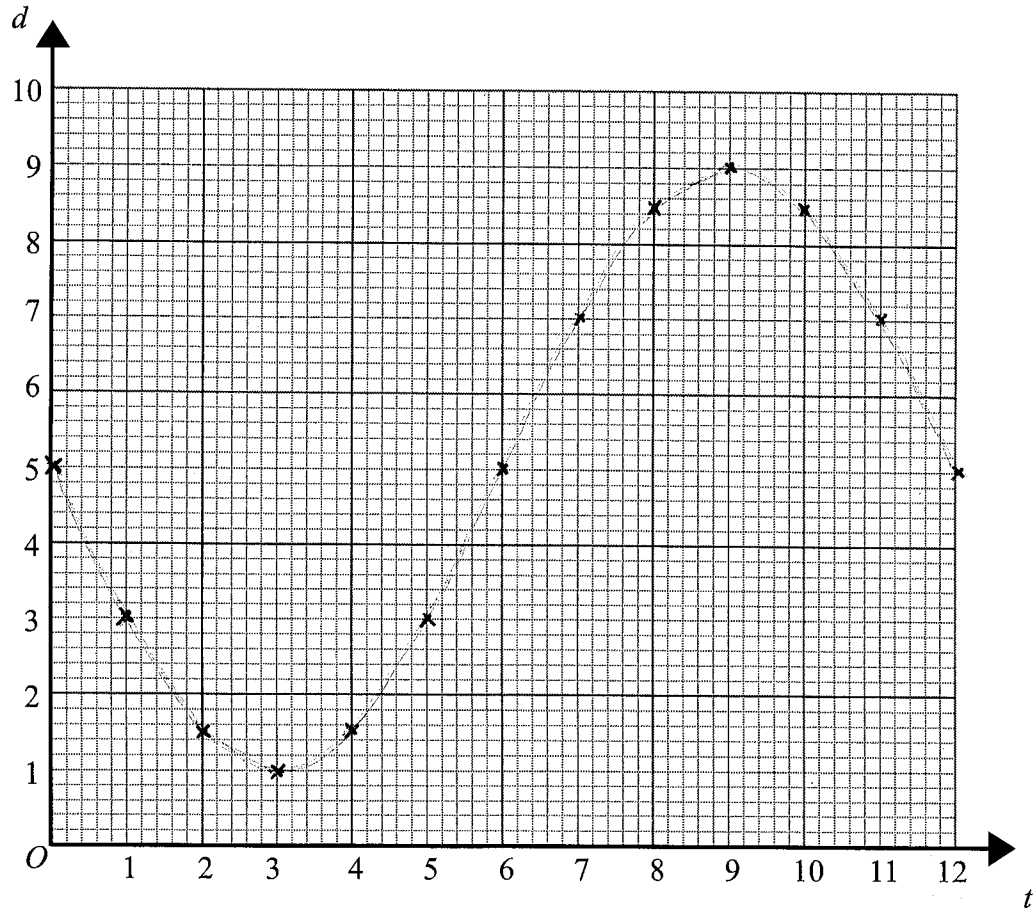
$\underline{\underline{p = 8}} \quad \underline{\underline{q = \frac{9}{4}}} \quad \underline{\underline{k = 27}}$

(Total for Question 8 is 6 marks)

9 The depth of water, d metres, at the entrance to a harbour is given by the formula: $d = 5 - 4\sin(30t)$, where t is the time in hours after midnight on one day.

(a) On the axes below, draw the graph of d against t for $0 \leq t \leq 12$ **(4)**

t	0	1	2	3	4	5	6	7	8	9	10	11	12
d	5	3	1.54	1	1.54	3	5	7	8.46	9	8.46	7	5



(b) Find the two values of t, where $0 \leq t \leq 24$, when the depth is least.

3 and 15

(2)

(Total for Question 9 is 6 marks)